አብነታት ልሳን ዐቢይ

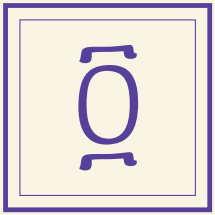
Large language models

Weekend 1: Representation and the transformer

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 16 October 2026 ፩፮፲፭

The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 4

Representation

Happy or unhappy?

The package was good and fine, so I am ___ happy unhappy

The package was late and broken, so I am ___ unhappy happy

Happy or unhappy?

The package was **good** and **fine**, so I am ___ → happy

The package was **late** and **broken**, so I am ___ → unhappy

Happy or unhappy?

The package was **good** and **fine**, so I am ___ → happy

The package was **late** and **broken**, so I am ___ → unhappy

You looked at a few telling words.

Happy or unhappy?

The package was **good** and **fine**, so I am ___ → happy

The package was **late** and **broken**, so I am ___ → unhappy

You looked at a few telling words.

A computer can do the same, if the words become numbers.

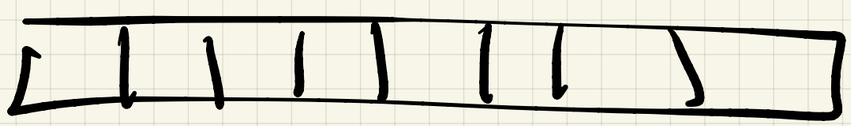
An invented corpus of reviews

Happy	Unhappy
The package was good and fine, so I am **happy**.	The package was late and broken, so I am **unhappy**.
The package was fine and the box was great, so I am **happy**.	The package was damaged and late, so I am **unhappy**.
The package was good, fine and great, so I am **happy**.	The package was fine but late and broken, so I am **unhappy**.

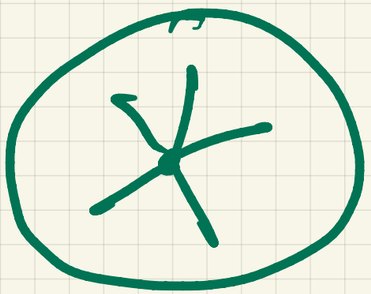
Modern LLMs

Transformer

prefix → Encoder → repr



Decoder



→ next word

An invented corpus of reviews

Happy	Unhappy
The package was good and fine, so I am **happy**.	The package was late and broken, so I am **unhappy**.
The package was fine and the box was great, so I am **happy**.	The package was damaged and late, so I am **unhappy**.
The package was good, fine and great, so I am **happy**.	The package was fine but late and broken, so I am **unhappy**.



*Happy reviews say **good**, **fine**, **great**. Unhappy ones say **late**, **broken**, **damaged**.*

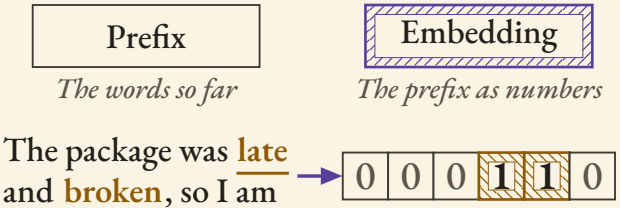
From a prefix to a guess

Prefix

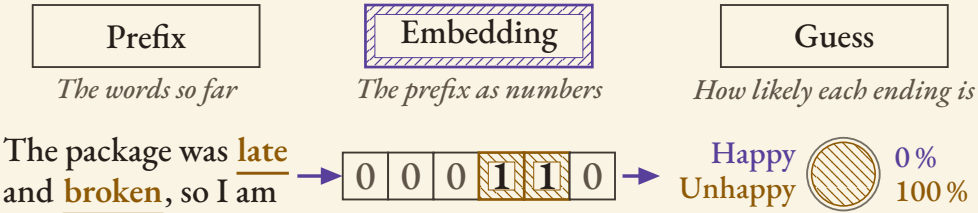
The words so far

The package was **late**
and **broken**, so I am

From a prefix to a guess



From a prefix to a guess



Count the words

The package was late and broken, so I am

The dictionary: three happy words and three unhappy words

good	fine	great	late	broken	damaged

Count the words

The package was late and broken, so I am

The dictionary: three happy words and three unhappy words

good	fine	great	late	broken	damaged
0	0	0	1	1	0

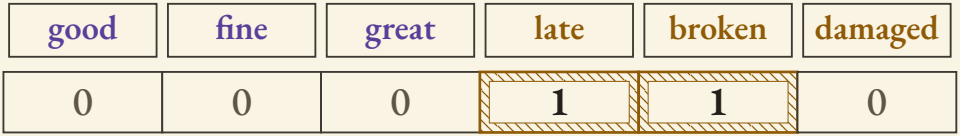
Every other word is ignored.

Encoding

Count the words

The package was late and broken, so I am

The dictionary: three happy words and three unhappy words

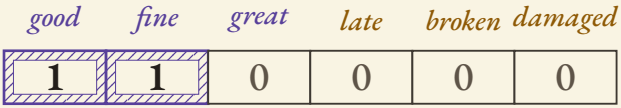


Every other word is ignored.

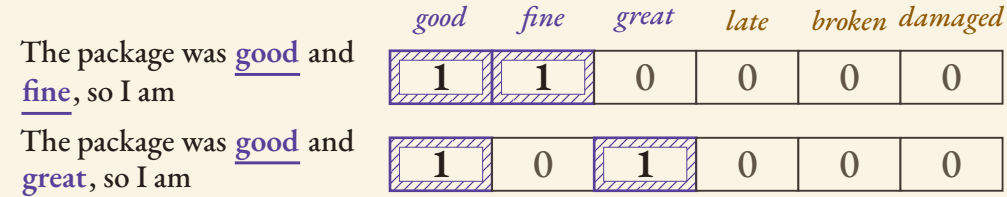
*This row of counts is the **embedding** of the prefix.*

Every prefix becomes a row

The package was good and fine, so I am

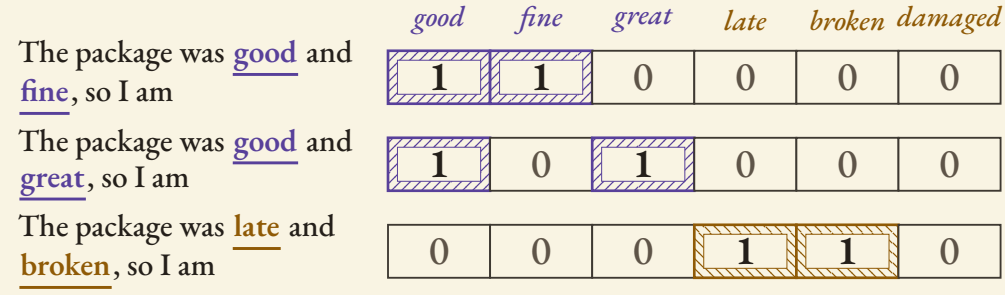


Every prefix becomes a row



Similar reviews get similar rows.

Every prefix becomes a row



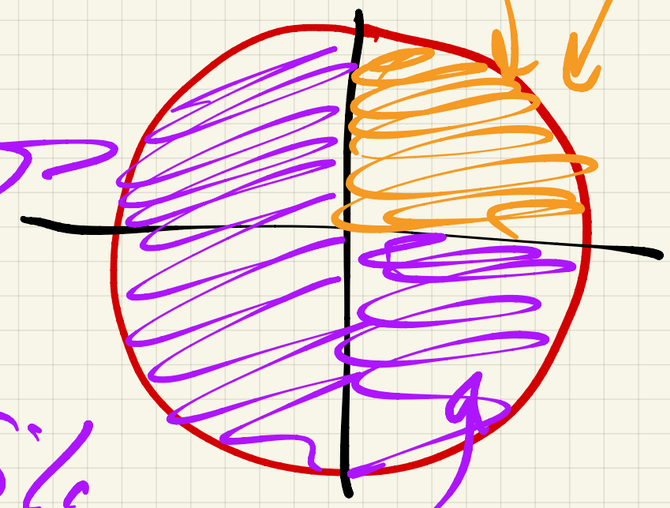
Similar reviews get similar rows.

A different kind of review gets a different row.

fine good great late broken damaged

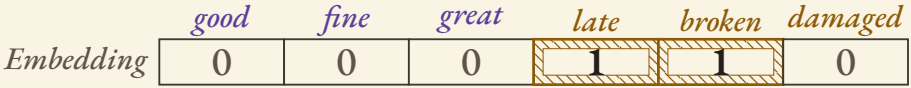
\+ + + − − −

2	0	1	0	1	0



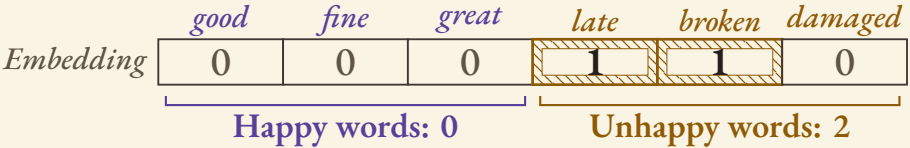
From counts to a guess

The package was **late** and **broken**, so I am



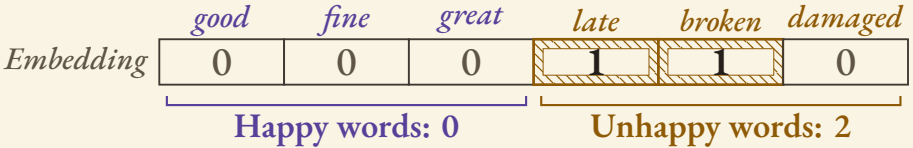
From counts to a guess

The package was **late** and **broken**, so I am



From counts to a guess

The package was late and broken, so I am

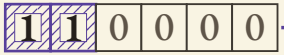


The whole pipeline

Prefix

The package was good and fine, so I am

Embedding

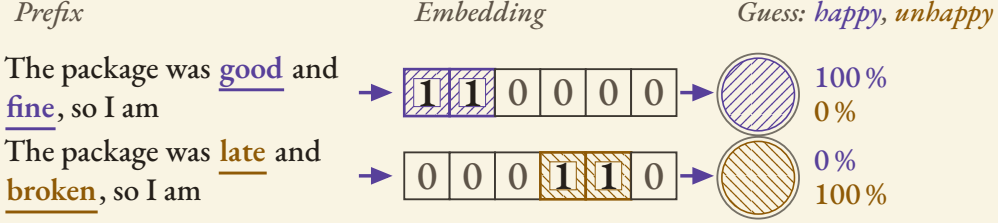


Guess: ***happy***, ***unhappy***

100 %
0 %

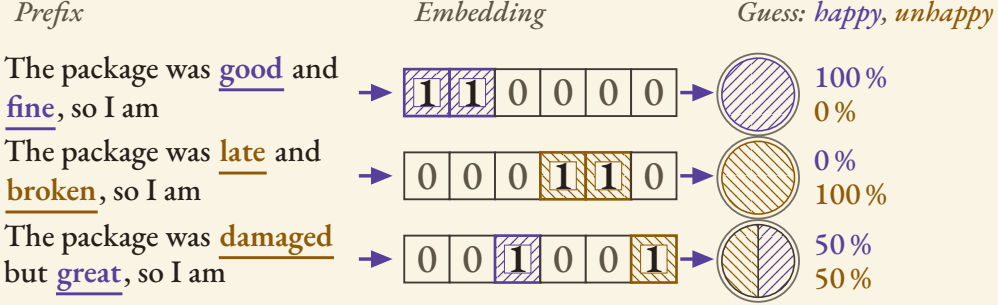
The cells follow the dictionary: ***good***, ***fine***, ***great***, ***late***, ***broken***, ***damaged***.

The whole pipeline



*The cells follow the dictionary: **good**, **fine**, **great**, **late**, **broken**, **damaged**.*

The whole pipeline

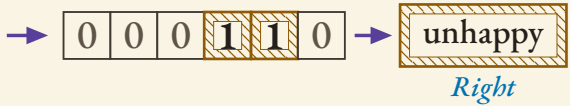


*The cells follow the dictionary: **good**, **fine**, **great**, **late**, **broken**, **damaged**.*

What counting misses

The review, with the ending it really has

The package was late and broken, so I am *unhappy*

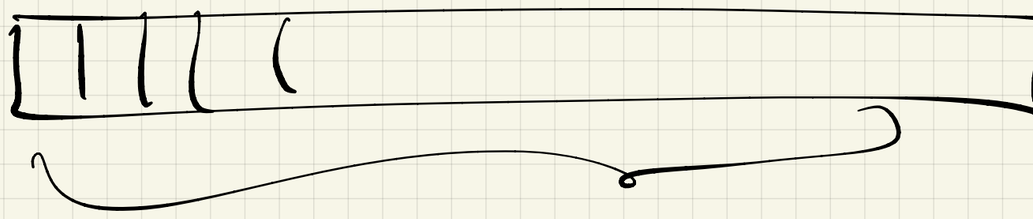


Ho

Modern LLMs

Encoding

Prefix



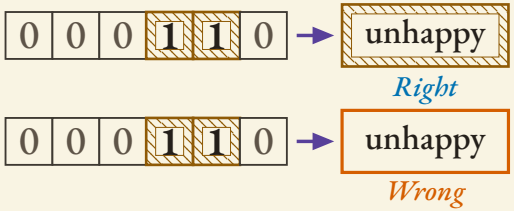
1000 measurements

What counting misses

The review, with the ending it really has

The package was **late** and **broken**, so I am *unhappy*

The package was not **late** and not **broken**, so I am *happy*



What counting misses

The review, with the ending it really has

The package was late and broken, so I am *unhappy*



The guess

unhappy

Right

The package was not late and not broken, so I am *happy*



unhappy

Wrong

The dictionary has no slot for the word not.

What counting misses

The review, with the ending it really has

The package was late and broken, so I am *unhappy*



The guess

unhappy

Right

The package was not late and not broken, so I am *happy*



unhappy

Wrong

The dictionary has no slot for the word not.

Counting forgets how the words combine.

What an embedding is



- **Embedding**: a text written as a row of numbers

What an embedding is



- **Embedding**: a text written as a row of numbers
- Here, one count for each word of a dictionary

What an embedding is



- **Embedding**: a text written as a row of numbers
- Here, one count for each word of a dictionary
- Happy words divided by all the words counted gives the chance of a happy ending

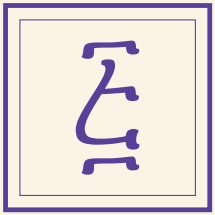
What an embedding is



- **Embedding**: a text written as a row of numbers
- Here, one count for each word of a dictionary
- Happy words divided by all the words counted gives the chance of a happy ending
- Real ones have thousands of numbers, learned by the model

The five Ge’ez games

	ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
	ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don’t understand.
	ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
	ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
	ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.



Section 5

Transformer

A language model is a table

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

A language model is a table

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

One row for each prefix.

A language model is a table

Prefix	Alice	Bob	plays	sings	chess	golf	tennis	END
START	0.6	0.4						
Alice			0.67	0.33				
Bob			1.0					
Alice plays						0.5	0.5	
Alice sings								1.0
Bob plays					0.5		0.5	
Alice plays golf								1.0
Alice plays tennis								1.0
Bob plays chess								1.0
Bob plays tennis								1.0

One row for each prefix.

A row holds the chance of each next word.

A row is a probability distribution

The row of the prefix **Alice**

Alice	Bob	plays	sings	chess	golf	tennis	END
0	0	**0.67**	**0.33**	0	0	0	0

A row is a probability distribution

The row of the prefix **Alice**

Alice	Bob	plays	sings	chess	golf	tennis	END
0	0	**0.67**	**0.33**	0	0	0	0



The same numbers, drawn as a roulette. A word with chance 0 gets no slice.

A row is a probability distribution

The row of the prefix **Alice**

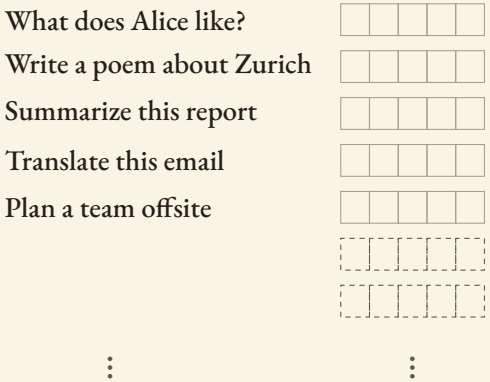
Alice	Bob	plays	sings	chess	golf	tennis	END
0	0	**0.67**	**0.33**	0	0	0	0



The same numbers, drawn as a roulette. A word with chance 0 gets no slice.

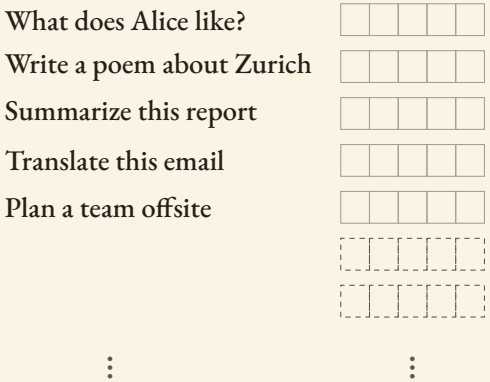
The slices add up to the whole wheel: a ***probability distribution****.*

The real table is too big



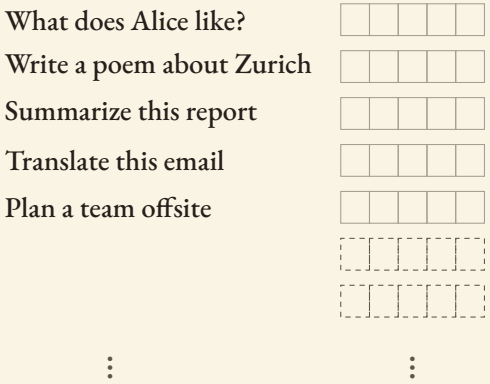
- One row for every possible prefix

The real table is too big



- One row for every possible prefix
- More rows than atoms in the universe

The real table is too big



- One row for every possible prefix
- More rows than atoms in the universe
- Too big for any computer

Compute each row when it is needed

The package was late and broken, so I am



Prefix	happy	unhappy
The package was good and fine, so I am		
The package was late and broken, so I am		
The package was damaged but great, so I am		

⋮

Compute each row when it is needed

The package was late and broken, so I am



Prefix	happy	unhappy
The package was good and fine, so I am		
The package was late and broken, so I am	0 %	100 %
The package was damaged but great, so I am		

⋮

Compute each row when it is needed

The package was late and broken, so I am



Prefix	happy	unhappy
The package was good and fine, so I am	100 %	0 %
The package was late and broken, so I am	0 %	100 %
The package was damaged but great, so I am	50 %	50 %

⋮

The table is never stored. Each row is computed from its prefix.

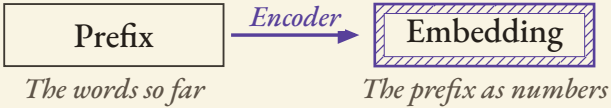
The recipe: encoder and decoder

Prefix

The words so far

Alice plays

The recipe: encoder and decoder



Alice plays

0.8	0.1	0.5	0.3	0.9	0.2

Invented numbers, for illustration

The recipe: encoder and decoder



The words so far — *The prefix as numbers* — *The chance of each next word*

Alice plays

0.8	0.1	0.5	0.3	0.9	0.2

Invented numbers, for illustration



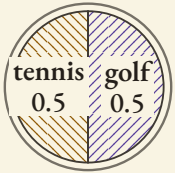
The recipe: encoder and decoder



Alice plays

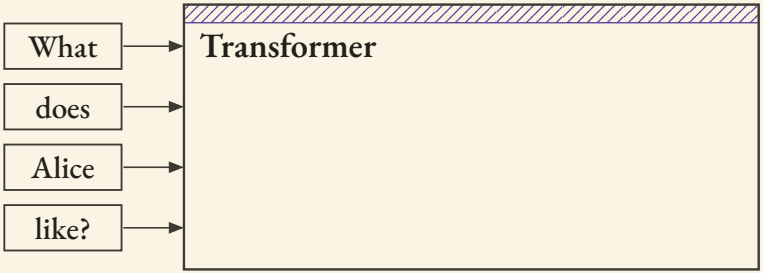
0.8	0.1	0.5	0.3	0.9	0.2

Invented numbers, for illustration

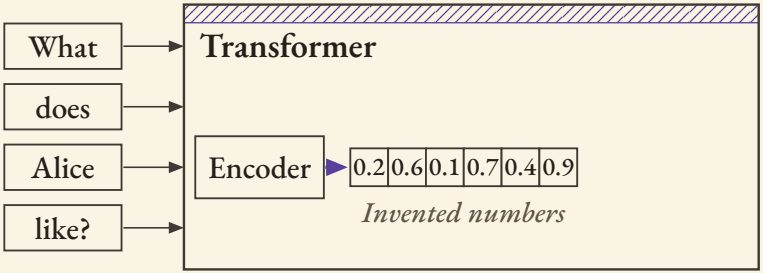


In Game 4 the encoder counted words, and the decoder divided.

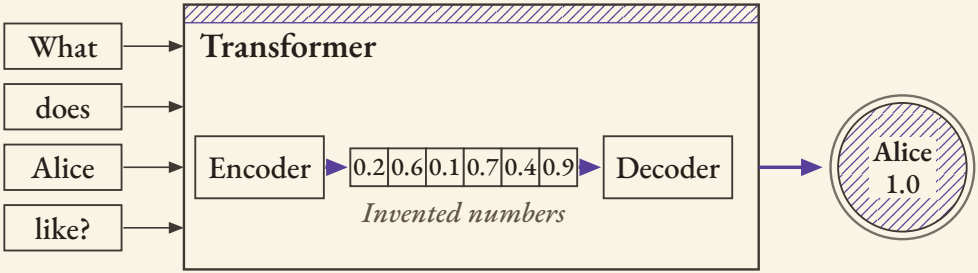
A transformer does both steps



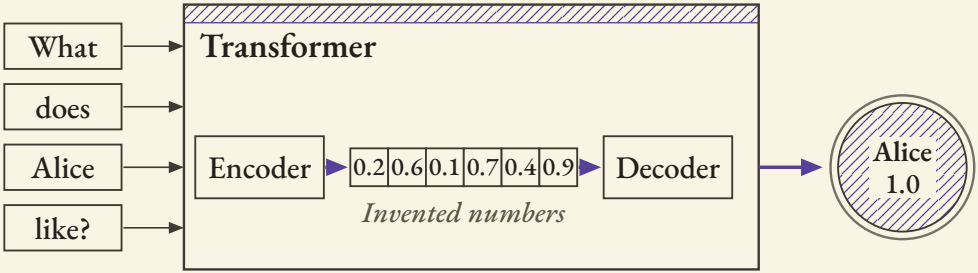
A transformer does both steps



A transformer does both steps

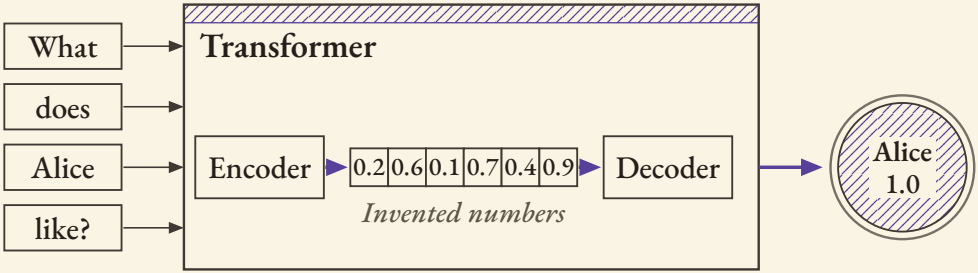


A transformer does both steps



It reads each word together with the words before it.

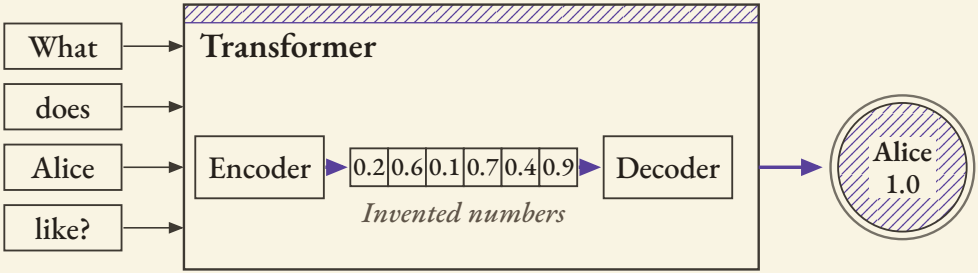
A transformer does both steps



It reads each word together with the words before it.

Its billions of numbers are learned from text.

A transformer does both steps



It reads each word together with the words before it.

Chat models do both steps in one stack, which is called decoder-only.

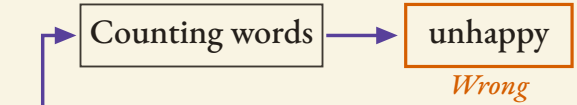
Its billions of numbers are learned from text.

A transformer reads how the words combine

The package was not
late and not broken, so
I am ___

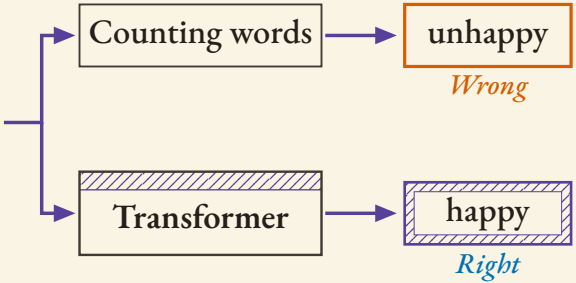
A transformer reads how the words combine

The package was not late and not broken, so I am ___



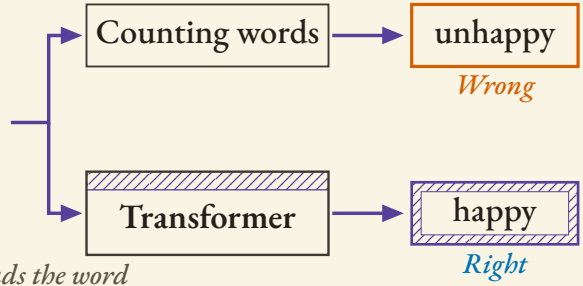
A transformer reads how the words combine

The package was not late and not broken, so I am ___



A transformer reads how the words combine

The package was not late and not broken, so I am ___



A trained transformer reads the word not together with the words it changes, and usually gets this right.

From our toys to a large language model

	Our toy, from Game 3	A large language model
Word pieces it knows	8	about 100 000

From our toys to a large language model

	Our toy, from Game 3	A large language model
Word pieces it knows	8	about 100 000
Numbers in an embedding	6	thousands

From our toys to a large language model

	Our toy, from Game 3	A large language model
Word pieces it knows	8	about 100 000
Numbers in an embedding	6	thousands
Numbers learned from text	14	billions

From our toys to a large language model

	Our toy, from Game 3	A large language model
Word pieces it knows	8	about 100 000
Numbers in an embedding	6	thousands
Numbers learned from text	14	billions
Text it learned from	14 words	trillions of word pieces

A prompt is a prefix

You

What does Alice like?

The model

Alice plays golf

A prompt is a prefix

You

What does Alice like?

The model

Alice plays golf

What does Alice like? **Alice plays golf** END

The prompt is the prefix

A prompt is a prefix

You

What does Alice like?

The model

Alice plays golf

What does Alice like? **Alice plays golf** END

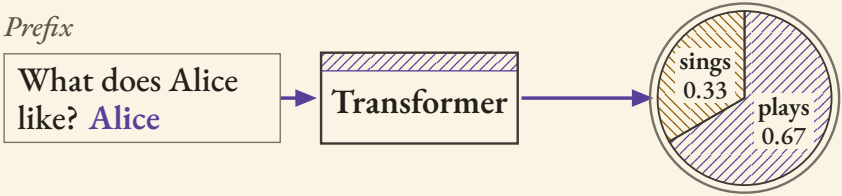
The prompt is the prefix *The answer, one word at a time*

One step: one new word

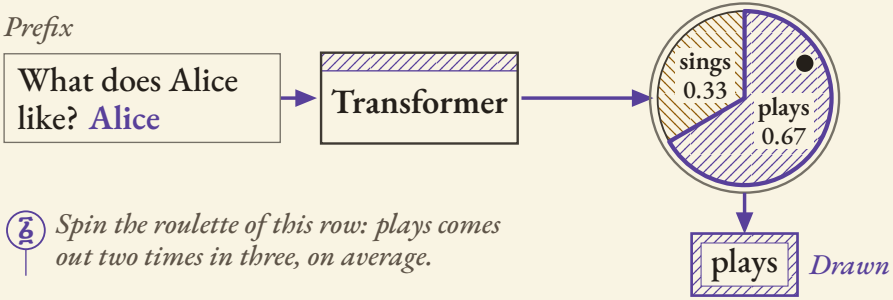
Prefix

What does Alice like? Alice

One step: one new word

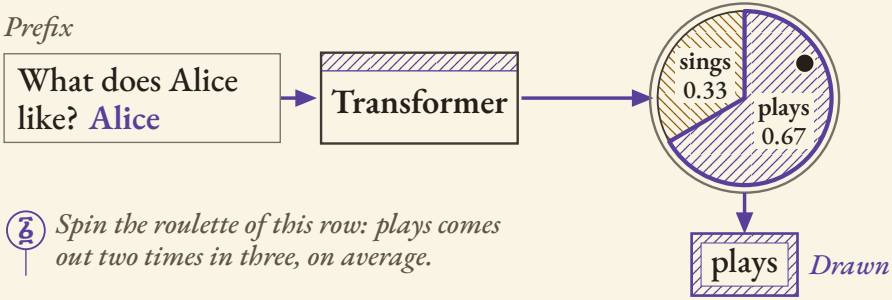


One step: one new word



Spin the roulette of this row: plays comes out two times in three, on average.

One step: one new word



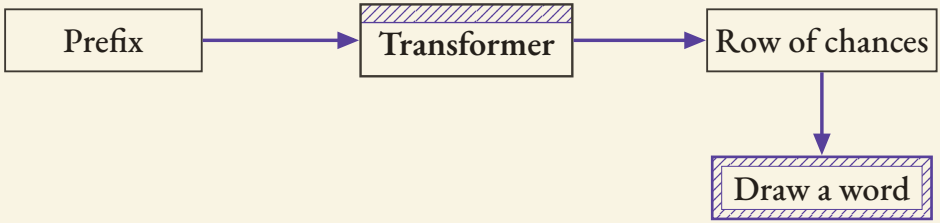
Spin the roulette of this row: plays comes out two times in three, on average.

One time in three, sings comes out instead.

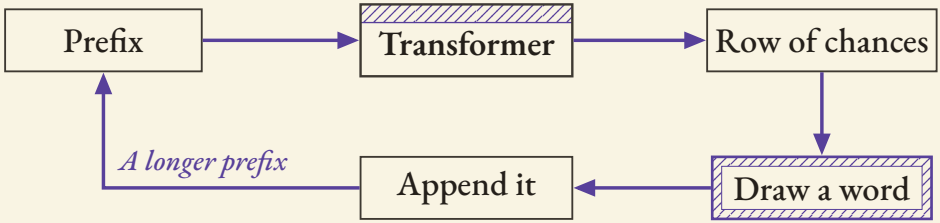
Append the word and repeat



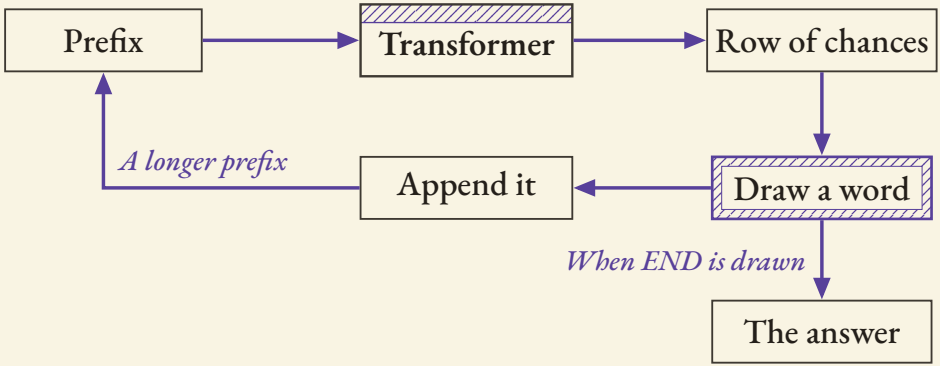
Append the word and repeat



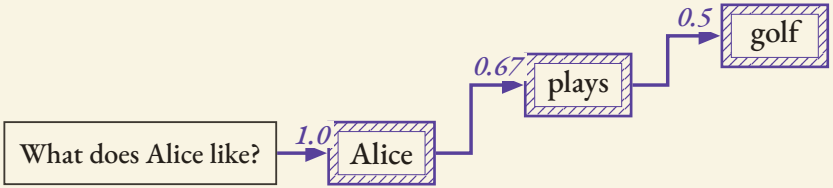
Append the word and repeat



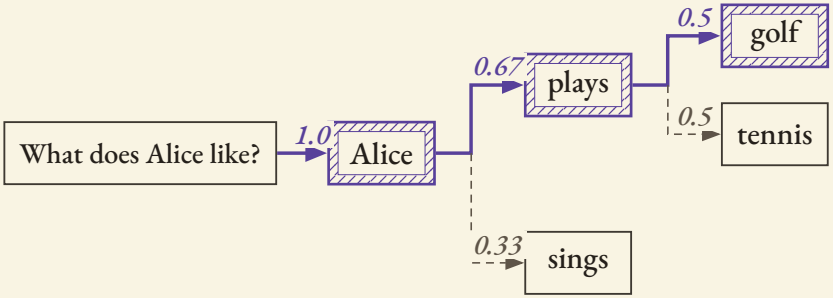
Append the word and repeat



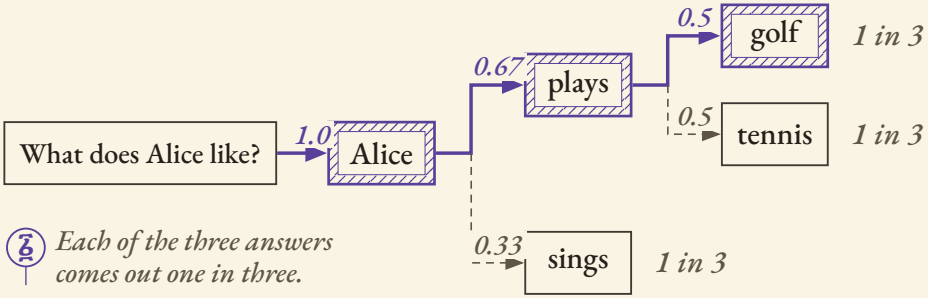
Ask again, get another answer



Ask again, get another answer

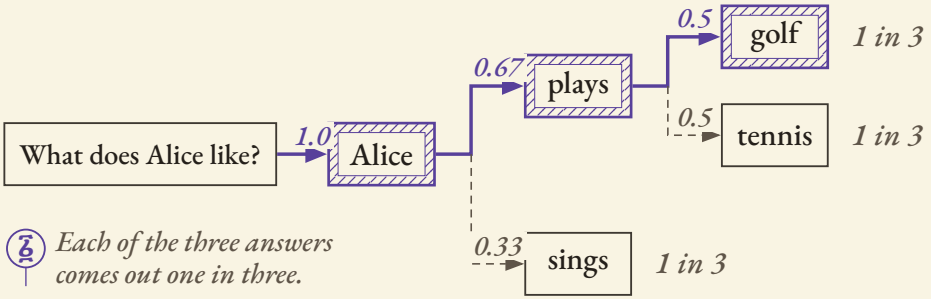


Ask again, get another answer



Each of the three answers comes out one in three.

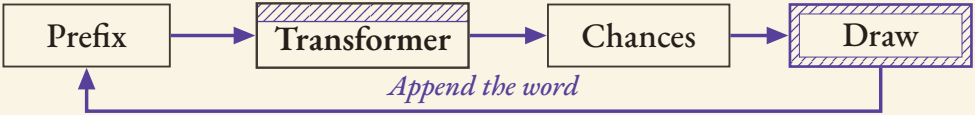
Ask again, get another answer



Each of the three answers comes out one in three.

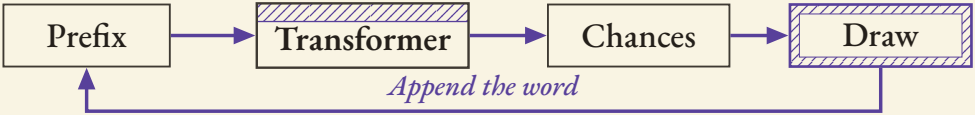
Ask again, and the answer can change.

How an LLM works



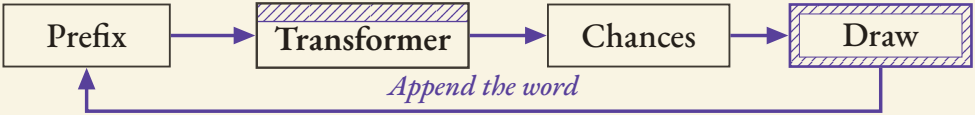
- A **prompt** is a prefix

How an LLM works



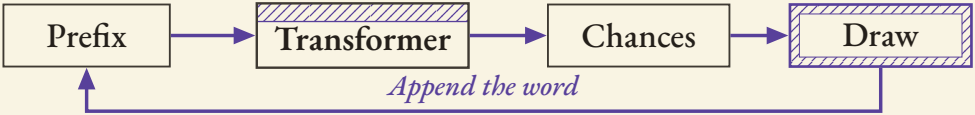
- A **prompt** is a prefix
- A **transformer** turns it into an embedding, then into a row of chances

How an LLM works



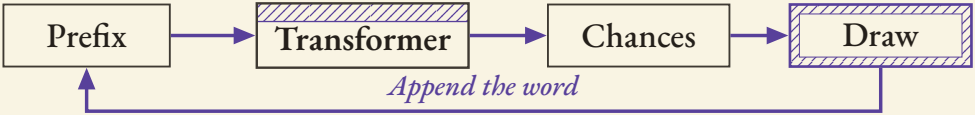
- A **prompt** is a prefix
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- One word is drawn by its chance and appended to the prefix

How an LLM works



- A **prompt** is a prefix
- A **transformer** turns it into an embedding, then into a row of chances
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- Repeat until END: the words drawn are the answer

How an LLM works



- A **prompt** is a prefix
- A **transformer** turns it into an embedding, then into a row of chances
- One word is drawn by its chance and appended to the prefix
- Repeat until END: the words drawn are the answer
- Chat models are then tuned on good answers and human ratings

Change the prefix, change the wheel

Prefix

Chances: happy, unhappy

The package was good and fine, so I am →



100 %
0 %

Change the prefix, change the wheel

Prefix

Chances: happy, unhappy

The package was good and fine, so I am →



100 %
0 %

The package was late and broken, so I am →



0 %
100 %

Change the prefix, change the wheel

Prefix

Chances: happy, unhappy

The package was **good** and **fine**, so I am →



100 %
0 %

The package was **late** and **broken**, so I am →



0 %
100 %

The package was **damaged** but **great**, so I am →



50 %
50 %

Change the prefix, change the wheel

Prefix

Chances: happy, unhappy

The package was **good** and **fine**, so I am →



100 %
0 %

The package was **late** and **broken**, so I am →



0 %
100 %

The package was **damaged** but **great**, so I am →



50 %
50 %

In a chatbot, the prefix is your prompt and the conversation so far.

Change the prefix, change the wheel

Prefix

Chances: happy, unhappy

The package was **good** and **fine**, so I am →



100 %
0 %

The package was **late** and **broken**, so I am →



0 %
100 %

The package was **damaged** but **great**, so I am →



50 %
50 %

In a chatbot, the prefix is your prompt and the conversation so far.

A prompt shapes the wheel. How to write a good one is the next lecture.

The roulette always lands on a word

Prefix with a name that the toy has never seen

What does Carol like? Carol plays

The toy table has no row for this prefix.

The roulette always lands on a word

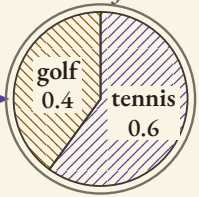
Prefix with a name that the toy has never seen

What does Carol like? Carol plays

The toy table has no row for this prefix.

Transformer

Invented chances, for illustration



The roulette always lands on a word

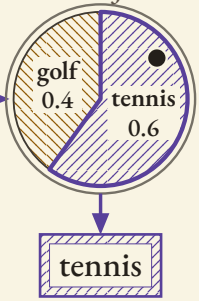
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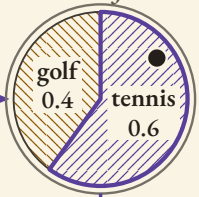
The roulette always lands on a word

Prefix with a name that the toy has never seen

What does Carol like? Carol plays

Transformer

Invented chances, for illustration



tennis

The toy table has no row for this prefix.

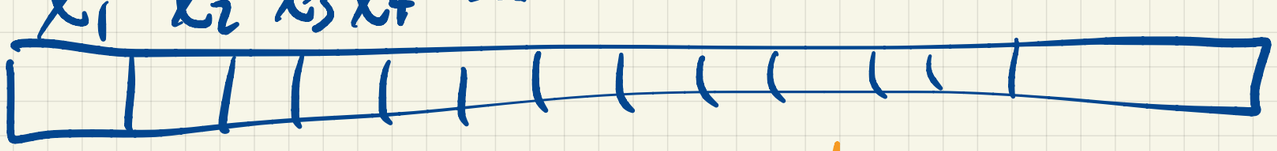
A real model always makes a row, even for a name it has never seen.

Prefix

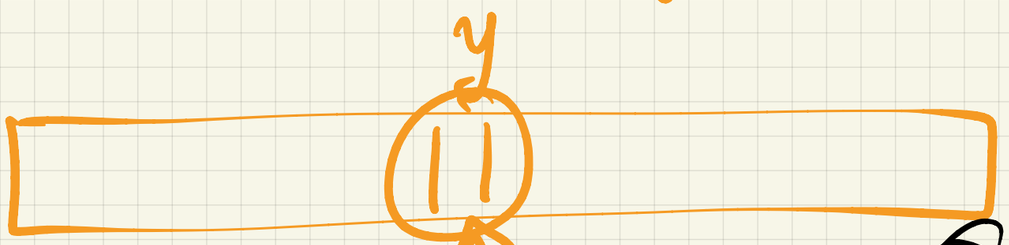
What Alice plays?

Encoding

$x_1\ x_2\ x_3\ x_4\ \cdots$



Encoder



Parameters

$y = w_1 x_1 + w_2 x_2 + \cdots$

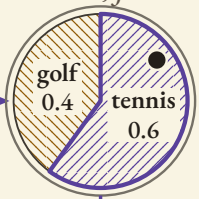
The roulette always lands on a word

Prefix with a name that the toy has never seen

What does Carol like? Carol plays

Transformer

Invented chances, for illustration



tennis

The toy table has no row for this prefix.

A real model always makes a row, even for a name it has never seen.

The model has no built-in step that checks a word against the world, so a fluent answer can be wrong. Check the facts that matter.

Summary

 ሥርዓት **Structure**	We can extract structure from unfamiliar text without knowing a single word.
 ምሳል **Imitation**	By spotting patterns, we can compose new sentences in a language we don't understand.
 ትንቢት **Prediction**	A language model picks the next word by probabilities conditioned on what came before.
 ምልክት **Representation**	A prefix becomes an embedding, and the embedding gives the chances of the next word.
 ቅንብር **Transformer**	An LLM computes the chances, draws a word, appends it, and repeats.